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Project Title	Probabilistic Customer Lifetime Value Estimation Using BG/NBD
Week	Week 1

WEEK-1 Technical Report

Task Name

Probabilistic Customer Lifetime Value Estimation Using BG/NBD

Abstract

Customer Lifetime Value (CLV) estimates the total value a customer brings to a business. The BG/NBD (Beta Geometric/Negative Binomial Distribution) model predicts future purchases using past transaction data. It helps businesses identify valuable customers, improve marketing strategies, and increase customer retention.

Introduction

Customer retention is more cost-effective than acquiring new customers. The BG/NBD model predicts future customer purchases based on historical buying behavior, making it useful for estimating CLV.

Theory

- **CLV:** Total expected profit from a customer.
- **BG/NBD:** A probabilistic model that predicts future purchases.
- **Inputs:** Frequency, Recency, Observation Time (T), and Monetary Value.

Objectives

- Estimate Customer Lifetime Value.
- Predict future purchases.
- Identify loyal customers.

- Improve customer retention.
- Support marketing decisions.

Methodology

1. Collect customer transaction data.
2. Clean and preprocess the data.
3. Calculate Frequency, Recency, and T.
4. Train the BG/NBD model.
5. Predict future purchases and CLV.
6. Analyze the results.

Dataset

Commonly uses the **Online Retail Dataset** with:

- Customer ID
- Invoice Date
- Quantity
- Unit Price
- Country

Likelihood

The BG/NBD model uses **Maximum Likelihood Estimation (MLE)** to estimate model parameters and predict customer purchasing behavior.

Mathematical Model

- Let:
- x = Frequency (repeat purchases)
- t_x = Recency (last purchase time)
- T = Observation period
- λ = Purchase rate
- p = Dropout probability

- **Formula**
- **Expected Future Purchases:**
- $E(X) = BG/NBD(x, t_x, T)$
- **Customer Lifetime Value (CLV):**
- $CLV = \text{Expected Purchases} \times \text{Average Monetary Value}$
- **Parameter Estimation:**
- Uses **Maximum Likelihood Estimation (MLE)** to estimate λ and p .
- **Purpose:** Predict future customer purchases and estimate Customer Lifetime Value (CLV)

Complexity Analysis

- **Time Complexity:** $O(n \times \text{iterations})$
- **Space Complexity:** $O(n)$

Results

- Predicts future customer purchases.
- Identifies high-value customers.
- Improves targeted marketing.
- Supports better business decisions.

Conclusion

The BG/NBD model is an effective method for estimating Customer Lifetime Value. It helps businesses predict customer behavior, improve retention, and increase long-term profitability.

References

1. Customer-Base Analysis in a Discrete-Time Noncontractual Setting
2. UCI Machine Learning Repository
3. Python for Data Analysis

